DGCA CPL / ATPL STUDY NOTES • RADIO NAVIGATION • CH 16

# ✈ Chapter 16: RNAV

Area Navigation Systems & FMS

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## 1. Introduction & Benefits

Area Navigation (RNAV) is a method of navigation that permits aircraft operations on any desired track within the coverage of ground-based navigation signals or within the limits of self-contained systems. Aircraft are no longer required to fly directly over ground beacons.

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Chapter

# 16

# Area Navigation Systems (RNAV)


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*Fig 16.1: RNAV chapter overview — navigation computer, CDU, HSI/CDI indicator system*


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### ✓ Benefits of RNAV

- Reduced distance, flight time, and fuel → lower costs
- Increased route capacity — more direct routes, parallel and bypass routes available
- Reduced vertical and horizontal separation criteria
- More flexible use of airspace — avoids congestion at beacon overhead positions

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## 2. RNAV Types & Levels

### 📡 B-RNAV vs P-RNAV

| Type | Accuracy | Mandate |
|---|---|---|
| **Basic RNAV (B-RNAV)** | **5 NM** on 95% of occasions | Mandatory for 30+ pax aircraft in Eurocontrol airspace |
| **Precision RNAV (P-RNAV)** | **1 NM** on 95% of occasions | Required for P-RNAV terminal routes |

### 📡 RNAV Dimensions

| Level | Capability |
|---|---|
| **2D RNAV** | Horizontal plane only (lateral) |
| **3D RNAV** | Horizontal + vertical (altitude guidance) |
| **4D RNAV** | Horizontal + vertical + timing (RTA — Required Time of Arrival) |

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## 3. Simple 2D RNAV System

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# 16 Area Navigation Systems (RNAV)





*Simple 2D RNAV — Navigation Computer, CDU, HSI/CDI indicator*

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### 📡 Simple 2D RNAV Components

- **Navigation Computer Unit**: processes VOR/DME inputs to compute position
- **Control and Display Unit (CDU)**: pilot enters waypoints; system computes tracking
- **Indicator**: CDI (Course Deviation Indicator) or HSI (Horizontal Situation Indicator) — both show deviation from computed track as if flying TO a waypoint

### ⚡ Simple 2D RNAV Limitations

- Position accuracy limited by VOR/DME coverage and accuracy
- Cannot operate beyond LOS of VOR/DME stations
- Only horizontal guidance (2D) — no VNAV capability
- Limited to pre-set waypoints; no automatic flight phase management

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## 4. Phantom Station (VORTAC RNAV)

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## Introduction

RNAV is defined as a method of navigation which permits aircraft operations **on any desired track** within the coverage of **station-referenced** navigation signal, **or** within the limits of **self-contained** navigation system.

An area navigation (RNAV) system is any system that allows the aircraft to be navigated to the required level of accuracy without the requirement to fly directly over ground based facilities.

The required accuracy is achieved by using some, or all, of the following inputs of information:

- VOR/DME
- ILS/MLS
- GNSS
- INS/IRS
- ADC
- Time

The information is processed within the system to give the most accurate and continuously updated position and the necessary outputs to provide the pilot with course, ETA etc.

## Benefits of RNAV

RNAV allows aircraft to take a more direct flight path appropriate to the route they are flying thereby improving the operating efficiency and helping in relieving congestion on the overcrowded airway system. To facilitate this, air traffic control centres have established RNAV routes which are more direct than the traditional airways system allows and do not require aircraft to regularly fly to the overhead of beacons. Hence the benefits are:

- A reduction in distance, flight time and fuel (and hence costs) by giving airlines and pilots greater flexibility and choice of routes.
- An increase in the present route capacity by making full use of the available airspace by providing more direct routes, parallel or dual routes and bypass routes for overflying aircraft in high density terminal areas.
- A reduction in vertical and horizontal separation criteria.

## Types and Levels of RNAV

There are two **types of RNAV:**

Basic RNAV (B-RNAV) which is required to give a position accuracy to **within 5 NM** on at least 95% of occasions. It is now mandatory for all aircraft carrying 30 passengers or more to have B-RNAV capability within Euro-control airspace.

Precision RNAV (P-RNAV) must be accurate to **within 1.0 NM** on at least 95% of occasions. P-RNAV routes are now being established in terminal airspace.





*Fig 16.1: Phantom station — computed waypoint positioned at offset from real VOR/DME*

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**Phantom Station Operation**

The RNAV computer calculates a **phantom VOR/DME station** (also called a waypoint) at a defined range and bearing from a real VOR/DME. The aircraft's CDI/HSI then indicates course and distance to this phantom station as if it were a real NAVAID.

1. Pilot programs phantom station position as range & bearing from VOR/DME (rho/theta)
2. Computer uses real VOR bearing (theta) and DME distance (rho) to calculate aircraft position
3. Computes QDM and distance to phantom station (waypoint)
4. Displays deviation from computed track on CDI/HSI

**Exam: Range Read-out on 2D RNAV** Q: Aircraft is 15 NM from VOR, phantom station 30 NM from VOR. Range readout = **20 NM** (from phantom, not VOR).

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## 5. Level 4 RNAV — Flight Management System (FMS)

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| WAYPOINT | STATION | FREQUENCY | RADIAL | DISTANCE | APPLICATION |
|---|---|---|---|---|---|
| 1 | DTY | 116.4 MHz | 067 | 42 | En route Nav. |
| 2 | POL | 112.1 MHz | 066 | 29 | En route Nav. |
| 3 | NEW | 114.25 MHz | 218 | 26 | En route Nav. |
| 4 | NEW | 114.25 MHz | 251 | 4 | Holding LOM |
| 5 | I-NC | 111.5 MHz | N/A | N/A | ILS |

## Principle of Operation of a Simple 2D RNAV System

Refer to *Figure 16.3*. The aircraft is flying from waypoint 1 (WP1) defined by DTY VOR/DME to waypoint 2 (WP2) defined by POL VOR/DME. As the aircraft arrives at WP1, POL is selected and the range and bearing measured (145(M)/104 NM). The RNAV knows its position with respect to POL and the pilot has already input waypoint 2 with respect to POL. The computer can now compute the track and distance from WP1 to WP2 (340(M)/102 NM) since it has two sides, the included angle and the orientation of magnetic north. The RNAV now continually computes the aircraft position with respect to POL and compares this position with the computed track to determine the cross-track error and the distance to go. Steering demands are fed to a CDI or HSI for the pilot to keep the aircraft on track and give continuous range read-out to WP2. It should be noted that on such a system the indications of deviation from track are in NM.

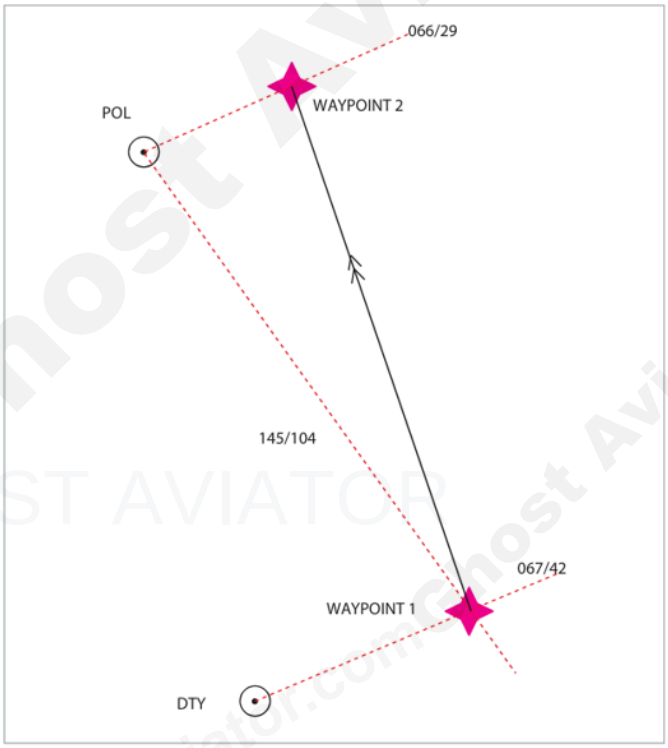



*Figure 16.3*



*Level 4 FMS — combines FMC, CDU, IRS/GPS inputs for full 4D navigation*

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### FMS Components (Boeing 737-400 / Similar Aircraft)

| Component | Function |
|---|---|
| **FMC (Flight Management Computer)** | Brain of FMS; processes navigation and performance data; calculates 4D trajectory |
| **CDU (Control & Display Unit)** | Pilot interface; alphanumeric keyboard + function keys + Line Select Keys (LSKs); scratchpad input |
| **IRS (Inertial Reference System)** | Self-contained position + attitude + groundspeed; degrades with time (drift) |
| **External references** | DME/DME, VOR/DME, GPS — FMC uses these to update IRS position |
| **Performance database** | Aircraft performance data for climb, cruise, descent optimization |
| **Navigation database** | Waypoints, airways, SIDs, STARs, airports — updated every **28 days** |

### ✓ FMS Capabilities (4D RNAV)

- LNAV (lateral navigation): autopilot follows computed lateral track
- VNAV (vertical navigation): computed climb/cruise/descent profiles
- RTA (Required Time of Arrival): timing function for 4D flight profiles
- Continuous optimization: adjusts speed/altitude for best fuel economy or ATC-assigned RTA
- Automatic DME/DME, VOR/DME or GPS position updates to IRS

## 6. CDU Pages

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There are three **levels of RNAV** capability:

- **2D RNAV** which relates to the capabilities in the **horizontal plane only**.
- **3D RNAV** indicates the addition of a guidance capability in the **vertical plane**.
- **4D RNAV** indicates the addition to 3D RNAV of a **timing function.**

## A Simple 2D RNAV System

The flight deck of a simple 2D RNAV system includes the following components:

- Navigation Computer Unit.
- Control and Display Unit (CDU).
- Indicator in the form of a:
  - Course Deviation Indicator (CDI) or
  - Horizontal Situation Indicator (HSI)

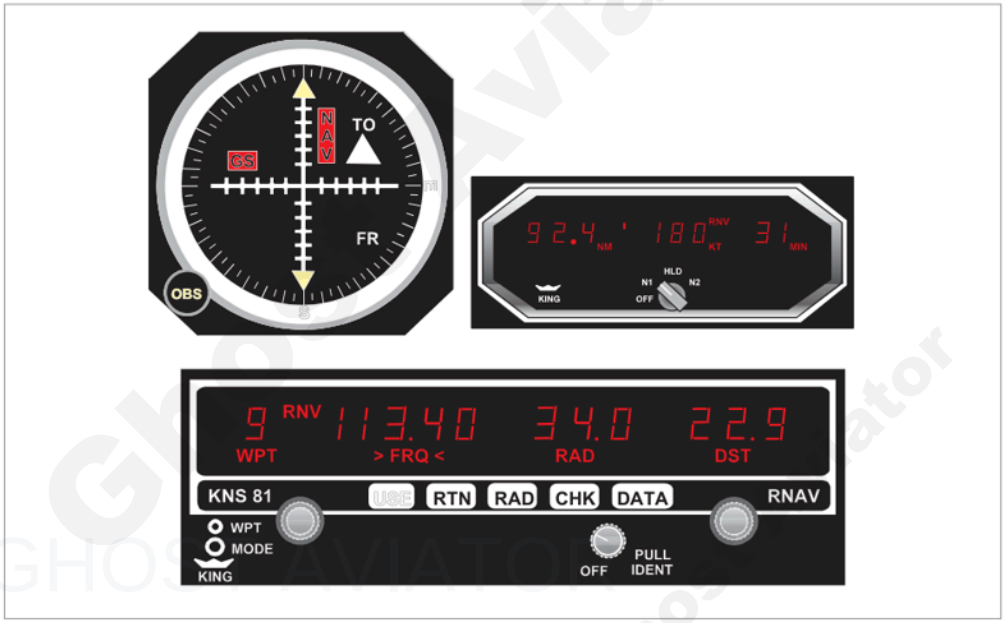



*Figure 16.1 VOR/DME RNAV Integrated Nav System*



*Fig 16.6: CDU layout — alphanumeric keyboard, function keys, line select keys (LSKs), scratchpad*

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## Operation of a Simple 2D RNAV System

A simple RNAV system uses rho/theta (range/bearing) to define position, which is derived from range and bearing information from VOR/DME stations. The pilot defines waypoints along the route to be flown as range and bearing from suitably located VOR/DME. Then the equipment, using the VOR/DME bearing and range, computes the QDM and distance to the waypoint and presents the information to the pilot on a CDI or HSI as if the waypoint itself is a VOR/DME station, hence these waypoints are known as phantom stations.

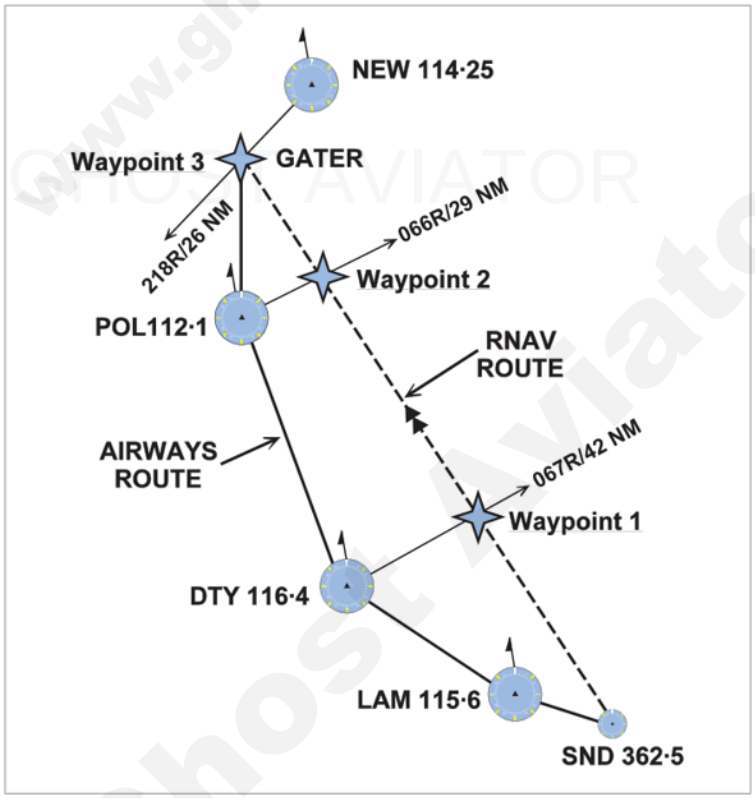



*Figure 16.2 An RNAV route & waypoints*

In the diagram the pilot has defined waypoints along the planned route from SND to NEW using available and sensibly placed VOR/DME.

Waypoints may be selected and programmed for:

- En route navigation.
- Initial approach fixes.
- Locator Outer Markers.
- ILS frequencies (when selected the instrumentation automatically reverts to ILS mode).

The following table shows the inputs that would be required for the above RNAV route.



*CDU pages: IDENT → POS INIT → RTE (pre-flight sequence)*



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### Pre-flight Initialization Sequence

| Page | Purpose | Key Inputs |
|---|---|---|
| **IDENT** | Verify aircraft/engine/database identity | Check nav data currency; update if required |
| **POS INIT** | Initialize IRS position & heading | Gate position or lat/long for IRS alignment; magnetic heading from standby compass |
| **RTE** | Input the route | Departure/destination airports, SID, airways, waypoints, runway |
| **CLB/CRZ/DES** | Review computed flight profiles | Check speeds, altitudes, restrictions, time/fuel predictions |

### CDU Waypoint Formats

- **5 alphanumerics**: named waypoints, VOR idents, airports (e.g. LOREL, TRN, EGLL)
- **SID/STAR**: up to 7 alphanumerics (e.g. TURN05)
- **Lat/Long**: N5000.0E00527.0 format
- **Place/Bearing/Distance (PBD)**: TRN250.0/76 (bearing from TRN, 76 NM)
- **Bearing/Bearing (BB)**: GOW167.0/TRN090.5 (intersection of bearings)

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## Limitations and Accuracy of Simple RNAV Systems

The beacons are selected by the pilot during the pre-flight planning and the pilot must ensure that each waypoint is within DOC of the VOR/DME designating that waypoint and of the VOR/DME designating the next waypoint.

Slant range error in DME must be considered in selecting facilities close to track.

The pilot must ensure that the information is correctly input into the CDU because the computer cannot recognize or rectify mistakes.

To avoid positional errors the aircraft must at all times be within the DOC of the in use facility. The accuracy of the fixing information is dependent on range and whether the VOR or DME element is predominant. If the VOR/DME is close to the planned track to/from the waypoint, the along track element will be most accurate. If the VOR/DME designating the way point is perpendicular to the track, the across track will be most accurate.





*CLB/CRZ/DES pages — VNAV profile management; shows ETA, fuel, restrictions, deviations*

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## Level 4 RNAV Systems

The area navigation function in modern passenger aircraft is carried out by a flight management computer (FMC) which also provides guidance and performance functions. The system outlined below is specific to the BOEING 737-800, but the principle is true for all aircraft.

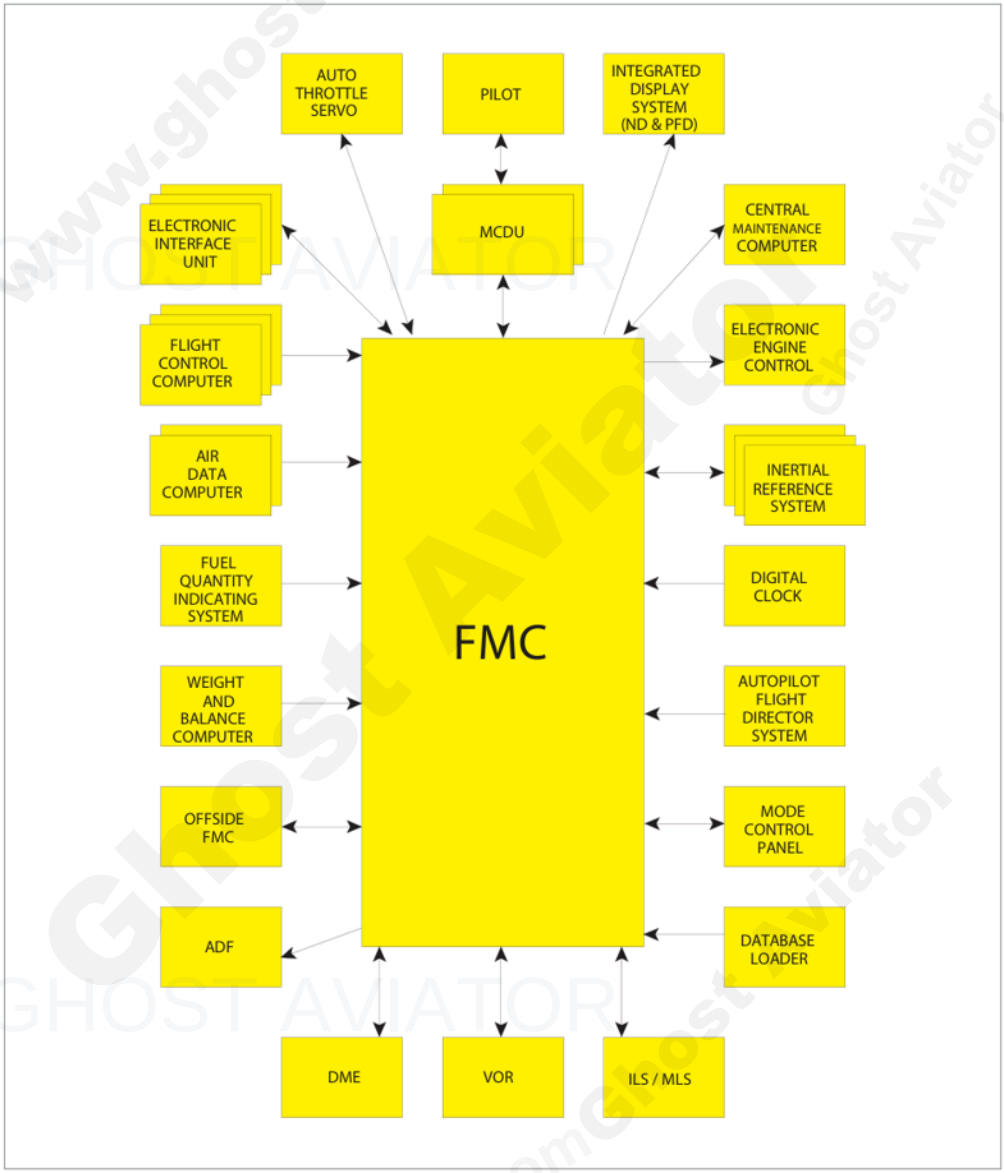



*Figure 16.4 FMS schematic*



*CDU cruise page — optimum/max altitude, step climb, wind entry, fuel prediction at destination*

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## 7. IRS/FMC Operation

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## Requirements for a 4D RNAV System

- Display present position in latitude/longitude or as distance/bearing to selected waypoint
- Select or enter the required flight plan through the control and display unit (CDU)
- Review and modify navigation data for any part of a flight plan at any stage of flight and store sufficient data to carry out the active flight plan
- Review, assemble, modify or verify a flight plan in flight, without affecting the guidance output
- Execute a modified flight plan only after positive action by the flight crew
- Where provided, assemble and verify an alternative flight plan without affecting the active flight plan
- Assemble a flight plan, either by identifier or by selection of individual waypoints from the database, or by creation of waypoints from the database, or by creation of waypoints defined by latitude/longitude, bearing/distance parameters or other parameters
- Assemble flight plans by joining routes or route segments
- Allow verification or adjustment of displayed position
- Provide automatic sequencing through waypoints with turn anticipation. Manual sequencing should also be provided to allow flight over, and return to, waypoints
- Display cross-track error on the CDU
- Provide time to waypoints on the CDU
- Execute a direct clearance to any waypoint
- Fly parallel tracks at the selected (offset distance offset mode must be clearly indicated)
- Purge previous radio updates
- Carry out RNAV holding procedures (when defined)
- Make available to the flight crew estimates of positional uncertainty, either as a quality factor or by reference to sensor differences from the computed position
- Conform to WGS-84 geodetic reference system
- Indicate navigation equipment failure





*IRS/FMC principle — triple IRS positions averaged, Kalman filtered, then updated by DME/DME or GPS*

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**Twin IRS, Twin FMC (Boeing 737)**

- Left FMC: uses left IRS as primary + right IRS cross-check; updated by external references
- Right FMC: uses right IRS as primary + left IRS cross-check
- If one FMC fails: other FMC takes over all functions
- If both FMCs fail: IRS information fed directly to EFIS; no automatic performance management

**Triple IRS, Twin FMC (Larger Aircraft)**

- All three IRS positions compared → gross error detection → averaged
- Averaged position compared with external reference (DME/DME, VOR/DME or GNSS)
- **Above 84° latitude**: de-couple IRS — secant of latitude approaches infinity → longitude calculation degrades rapidly

⚡ **FMC Position Accuracy Over Time** IRS drift accumulates during flight → FMC most accurate at departure (IRS aligned), least accurate toward end of long flight if no external DME/GPS updates.

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## 8. Kalman Filtering

**Kalman Filtering**

A mathematical algorithm that optimally combines multiple sensor inputs (IRS position/velocity + external DME/GPS fixes) with their known error characteristics to produce the best estimate of current position and velocity. It:

- Weights each input inversely proportional to its error variance
- Smooths noisy GPS/DME updates against more stable (but drifting) IRS data
- Produces a position estimate with lower error than any individual input
- Continuously updates the position estimate as new sensor data arrives

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## 9. RNAV Summary

| Parameter | Detail |
|---|---|
| Definition | Navigation on any desired track within NAVAID coverage or self-contained system limits |
| B-RNAV | ±5 NM, 95% of occasions; mandatory 30+ pax in Eurocontrol |
| P-RNAV | ±1 NM, 95% of occasions; terminal routes |
| 2D RNAV | Lateral only; uses VOR/DME rho/theta; phantom station |
| 3D RNAV | + vertical guidance (VNAV) |
| 4D RNAV | + timing (RTA) |
| FMS | FMC + CDU + IRS + Nav/Perf Database; full 4D capability |
| Nav database update | Every 28 days (AIRAC cycle) |
| CDU sequence | IDENT → POS INIT → RTE → CLB/CRZ/DES |
| IRS de-coupling | Above 84° latitude (longitude error growth) |
| Kalman filtering | Optimally combines IRS + external references |

## 10. Practice Questions

**Q1. The accuracy required of a precision area navigation system is:**

**(a)** 0.25 NM

**(b)** 2 NM

**(c)** 1 NM

**(d)** 0.5 NM

> **Answer: (c)**
> P-RNAV requires position accuracy of **1 NM** on 95% of occasions. B-RNAV requires 5 NM.

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**Q2. A basic 2D RNAV system will determine tracking information from:**

**(a)** twin DME

**(b)** VOR/DME

**(c)** twin VOR

**(d)** any of the above

**Answer: (b)**
A basic 2D RNAV system uses **VOR/DME** (rho/theta — range and bearing) to define position and compute track to waypoints.

**Q3. Aircraft on track WP1→WP2 (45 NM). Aircraft 20 NM from phantom station, phantom 270°/30 NM from VOR/DME. Aircraft 15 NM from VOR. Range read-out:**

**(a)** 15 NM

**(b)** 20 NM

**(c)** 25 NM

**(d)** 30 NM

**Answer: (b)**
The DME display shows distance to the **phantom station (waypoint)**, not the real VOR/DME. The aircraft is 20 NM from the phantom station → **20 NM**.

**Q4. The sequence of displays accessed on FMC initialization is:**

**(a)** POS INIT, IDENT, RTE

**(b)** IDENT, RTE, POS INIT

**(c)** IDENT, POS INIT, RTE

**(d)** POS INIT, RTE, IDENT

**Answer: (c)**
Correct pre-flight sequence: **IDENT → POS INIT → RTE**. IDENT first to verify database, then position initialization, then route entry.

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**Q5. The IRS position can be updated:**

**(a)** on the ground only

**(b)** at designated positions en route and on the ground

**(c)** on the ground and overhead VOR/DME

**(d)** at selected waypoints and on the ground

**Answer: (a)**
The IRS position can only be initialized on the **ground**. In flight, the FMC uses Kalman filtering with external references (DME/DME, GPS) to correct IRS drift, but the IRS itself is not manually updated in flight.

**Q6. Best action to insert accurate IRS position using POS INIT page (Appendix A):**

**(a)** 3R then 4R

**(b)** 2R then 4R

**(c)** 4R then 3R

**(d)** 3L then 4R

**Answer: (a)**
Select **3R** to transfer gate position to scratchpad, then **4R** to insert into IRS position field — this uses the most accurate database position.

**Q7. The position used by the FMC in the B737-400 is:**

**(a)** an average of the two IRS positions

**(b)** an average of the two IRS positions smoothed by Kalman filtering

**(c)** taken from the selected IRS, smoothed by Kalman filtering and updated to external reference

**(d)** generated from external reference, updated by IRS as part of Kalman filtering

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**Answer: (c)**
The FMC uses the position from the **selected IRS**, processed by **Kalman filtering** and updated by external references (DME/DME, VOR/DME, GPS).

**Q8. The FMC position will be at its most inaccurate:**

**(a)** on take-off

**(b)** at TOC

**(c)** at TOD

**(d)** on final approach

**Answer: (c)**
IRS drift accumulates over time. The position is most inaccurate at **Top of Descent (TOD)** — the end of a long cruise, furthest from the last IRS alignment. FMC tries to correct with DME/GPS updates.

**Q9. Positions input to FMC using maximum 5 alphanumerics:**

**(a)** SIDs & STARs, reporting points and airways

**(b)** Navigation facilities, reporting points and airways

**(c)** SIDs & STARs and latitude/longitude

**(d)** Latitude/longitude, reporting points and airways

**Answer: (b)**
Max 5 alphanumerics: **navigation facilities (NAVAIDS), reporting points, and airways designators**. SIDs/STARs use up to 7. Lat/long uses special format N5000.0E00527.0.

**Q10. The FMC navigation database can be accessed by pilots:**

**(a)** to update the database

**(b)** to read information only

**(c)** to change between 28-day updates

**(d)** to change data for sector requirements

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**Answer: (b)**
Pilots can **read (interrogate) the database only**. The database cannot be modified by pilots; it is updated by maintenance/data service providers every 28 days.

**Q11. Above 84° latitude, a twin FMS/triple IRS system goes to de-coupled operations because:**

**(a)** to prevent error messages as IRS longitudes show large differences

**(b)** to ease pilot workload

**(c)** to improve system accuracy

**(d)** because magnetic variation changes rapidly

**Answer: (a)**
Above 84° latitude, the secant of latitude increases rapidly. Longitude calculations from the two IRS systems show increasingly different values → **de-coupled mode** prevents false gross error messages.

**Q12. Maximum range at which VOR bearing information is used by the B737-400 FMC:**

**(a)** 160 NM

**(b)** 180 NM

**(c)** 200 NM

**(d)** 250 NM

**Answer: (b)**
The B737-400 FMC uses VOR bearing up to **180 NM** from the station. Beyond this range, bearing accuracy is insufficient for Kalman filtering updates.

**Q13. Which of the following is NOT a waypoint definition format for the CDU:**

**(a)** 5-letter alphanumeric

**(b)** Lat/Long

**(c)** range and bearing from NAVAID

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**(d)** bearing from two NAVAIDs

> **Answer: (d)**
> The bearing/bearing (BB) format (e.g. GOW167.0/TRN090.5) IS a valid waypoint format. The format NOT available is bearing from two NAVAIDs as a simple format — the CDU uses only the BB format with specific syntax.

**Q14. Correct format to input position 50°N 005°27'E to the CDU:**

**(a)** 5000.0N00527.0E

**(b)** N50E00527

**(c)** N5000.0E00527.0

**(d)** N5000E00527

> **Answer: (c)**
> CDU latitude/longitude format: **N5000.0E00527.0** — hemisphere prefix, degrees and decimal minutes, with decimal point mandatory.

**Q15. The period of validity of the FMC navigation database is:**

**(a)** 28 days

**(b)** 1 month

**(c)** determined by national authority (28 to 91 days)

**(d)** 91 days

> **Answer: (a)**
> The FMC navigation database is valid for **28 days** (one AIRAC cycle). It must be updated every 28 days to remain current.

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